

Inspection Report V-1201 — Shell Thickness Survey March 2029

1. Equipment Details

Equipment: V-1201 Crude Feed Surge Drum. Design pressure 18.0 barg. Design temperature 180 degC. Material SA-516 Gr.70. Nominal shell thickness 14.0 mm. Inside radius 1200 mm. Joint efficiency 0.85. Commissioned 2011.

2. Inspection Method

Ultrasonic thickness measurement at six established condition monitoring locations, performed in accordance with API 510. Instrument: Olympus 38DL Plus, calibrated 2029-02-28. Surface prepared by wire brushing.

3. Thickness Survey

CML	Location	2019 (mm)	2023 (mm)	2029 (mm)	t-min (mm)
CML-01	Top head	14.20	13.85	13.40	9.50
CML-02	Shell course 1	14.10	13.20	12.15	9.50
CML-03	Shell course 2	14.05	13.00	11.80	9.50
CML-04	Bottom shell	13.90	12.50	9.20	8.00
CML-05	Bottom head	14.00	13.60	13.05	8.00
CML-06	Nozzle N1 neck	11.50	11.10	10.60	7.00

4. Findings

CML-04 on the bottom shell shows the highest metal loss, measuring 9.20 mm against 12.50 mm recorded in 2023. This represents 3.30 mm of loss over six years. Localised pitting was observed adjacent to the bottom drain nozzle, consistent with water accumulation and under-deposit corrosion.

5. Assessment

The corrosion rate at CML-04 is the controlling value for the vessel. Against a minimum required thickness of 8.00 mm, the remaining thickness available for corrosion is 1.20 mm. Remaining life should be recalculated and the next inspection interval set at no more than half that value.

6. Recommendations

Increase inspection frequency at CML-04 to annual. Install a corrosion coupon in the bottom draw-off line. Review the water draw-off procedure with Operations. Consider internal coating at the next turnaround.

7. Sign-off

Inspected by: S. Nair, Senior Inspection Engineer, Level II UT. Reviewed by: M. Devadiga, Maintenance Head. Date: 2029-03-06.